

LIU HENGZHI

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EDUCATION

National University of Singapore **2026 – Present**
Bachelor of Computing in Computer Science (Expected to graduate in 3 years) *Singapore*

Temasek Polytechnic **2023 – 2026**
Diploma in Biomedical Engineering with Merit (GPA: 3.91/4.00) *Singapore*

- Polytechnic Scholarship (Engineering); Course Bronze Medalist; Director's List
- Focused on embedded systems, hardware-software co-design, and applied engineering research

SKILLS

Programming Languages: Python, C++, C, SQL (MySQL)

Systems & AI Infra: LLM Agents, System Design Basics, Embedded Systems, Real-time Systems (FreeRTOS), Firmware Development

Tools & Frameworks: Git/GitHub, Linux, FastAPI, PostgreSQL, AWS (Aurora/IAM), CAD, Adobe Suite

TECHNICAL PROJECTS

DTx Cognitive SaaS (ADHD Therapeutics)

Full-Stack Developer *React, FastAPI, AWS Aurora, PostgreSQL, LLM Agents*

- Ensured clinical ADHD assessment precision by building a B2B SaaS platform featuring adaptive pacing calibrated to patient severity.
- Reduced data upload lag by implementing local telemetry batching and securing connections via AWS token authentication.
- Built an autonomous AI Copilot Agent via Gemini API that automatically processes patient behaviors and caches multilingual reports to avoid API rate limits.

Smart Mahjong Play-Board

Independent Systems Developer *C++, ESP32, PCB Design, Embedded Firmware*

- Developed ESP32-based smart play-board hardware with custom PCB integration.
- Achieved lag-free, real-time physical-to-digital state updates by writing optimized ESP32 C++ firmware.

WORK EXPERIENCE

National University of Singapore (iHealthtech)
Research Assistant Intern

Mar 2025 – Jul 2025
Singapore

- Generated high-quality experimental datasets for biomedical research by setting up microscale testing workflows.
- Ensured data integrity across all testing pipelines by performing comprehensive validation and validation checks on hardware instruments.
- Guided project decision-making by converting complex experimental sensor data into clear, actionable reports.
- Maintained continuous lab operations by optimizing the allocation of shared equipment resources.
- Improved data traceability by establishing structured digital documentation standards for all experimental records.

UNIVERSITY ENGAGEMENT

- Incoming Undergraduate Student at School of Computing, National University of Singapore.
- Active engagement in computing community frameworks and prospective engineering alignment programs.

CO-CURRICULAR ACTIVITIES

WorldSkills Singapore (Additive Manufacturing)
• 1st Place (Qualifying Round); demonstrated advanced CAD modeling, additive manufacturing optimization, and engineering precision design

2024

Electronic Design & Robotics Competitions
• Developed embedded logic systems and real-time control algorithms for robotics and circuit design challenges

2024

ADDITIONAL INFORMATION

- **Leadership:** Temasek LEAD & ENGenius Excellence Programs alumnus
- **Languages:** English (Professional), Mandarin (Native)
- **Interests:** Distributed Systems, AI Infrastructure, Embedded Systems, Scientific Computing